

Marmara University

Electric-Electronic Engineering

Digital Design

CSE3017

Assignment 1

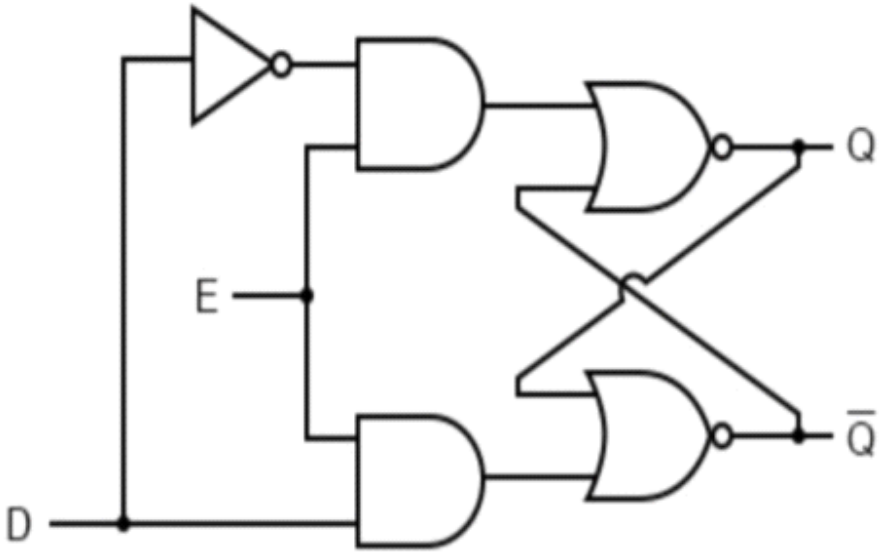
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In this assignment we will write verilog HDL code for D-latch circuit. D-latch circuit is given below.



The truth table for this circuit is like this:

E	D	Q	Q’
0	0	Latch	Latch
0	1	Latch	Latch
1	0	0	1
1	1	1	0

Finally we can start to write verilog HDL code for this circuit:

Firstly I have started with writing codes for Gates

AND GATE VERİLOG HDL CODE:

```
module andGate(output c, input a, b);
        assign c = a&b;
endmodule
```

NOT GATE VERİLOG HDL CODE:

```
module notGate(output a, input b);
```

```
    assign a = ~b;
endmodule
```

NOR GATE VERİLOG HDL CODE:

```
module norGate(output c, input a, b);
    assign c = ~(a|b);
endmodule
```

- **Now let's see the main code:**

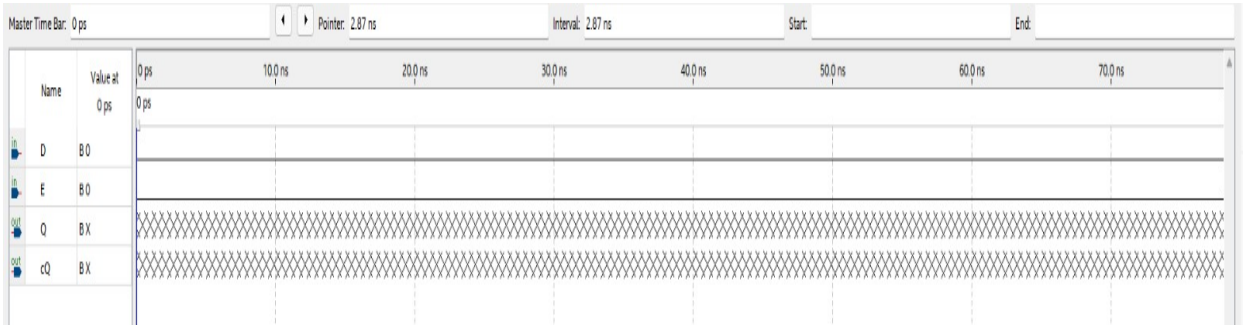
D-LATCH CIRCUIT VERİLOG HDL CODE:

```
module dLatch(output Q, cQ, input D, E);
    wire a1, a2, b1;
    notGate(a1, D);
    andGate(a2, a1, E);
    andGate(b1, D, E);
    norGate(Q, a2, cQ);
    norGate(cQ, b1, Q);
endmodule
```

Simulation (D-latch):

Also we tested and simulated this code on multisim:

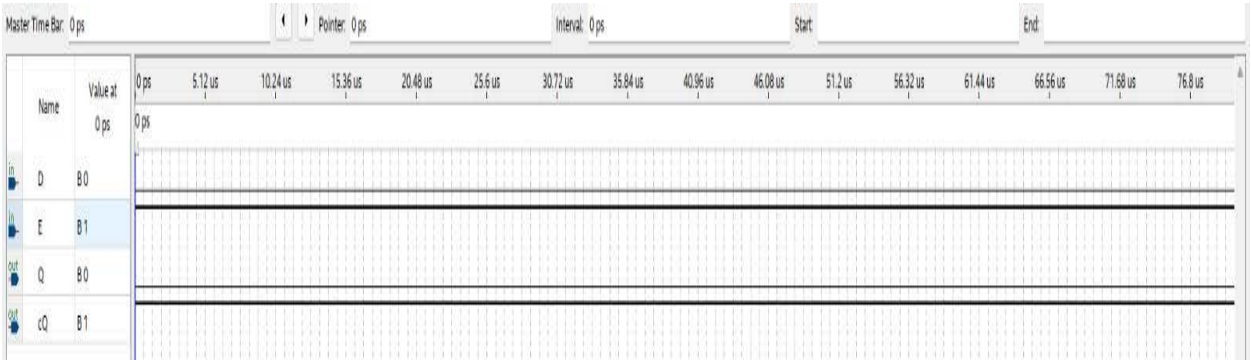
For D = 0, E = 0



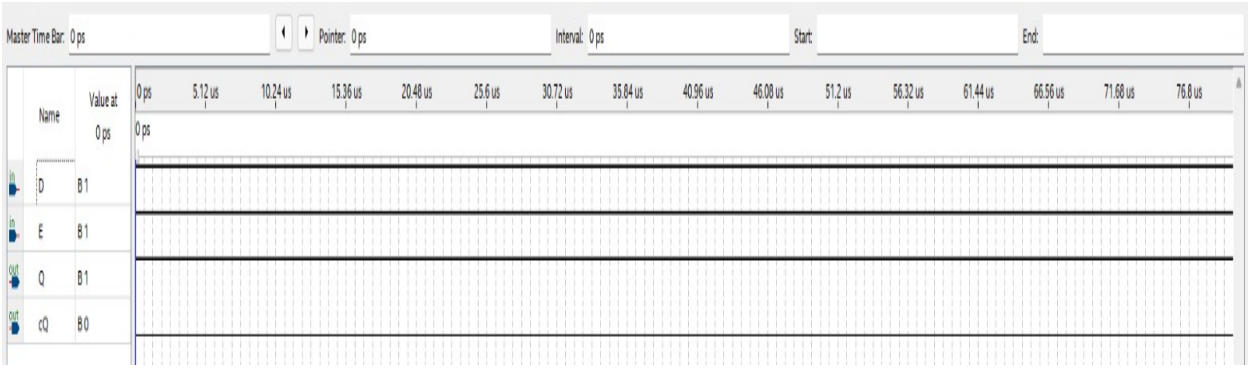
For D = 1, E = 0



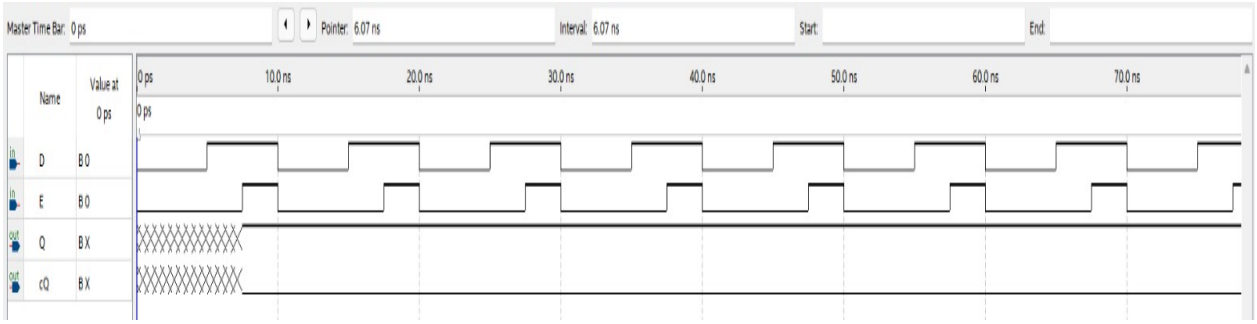
For D = 0, E = 1



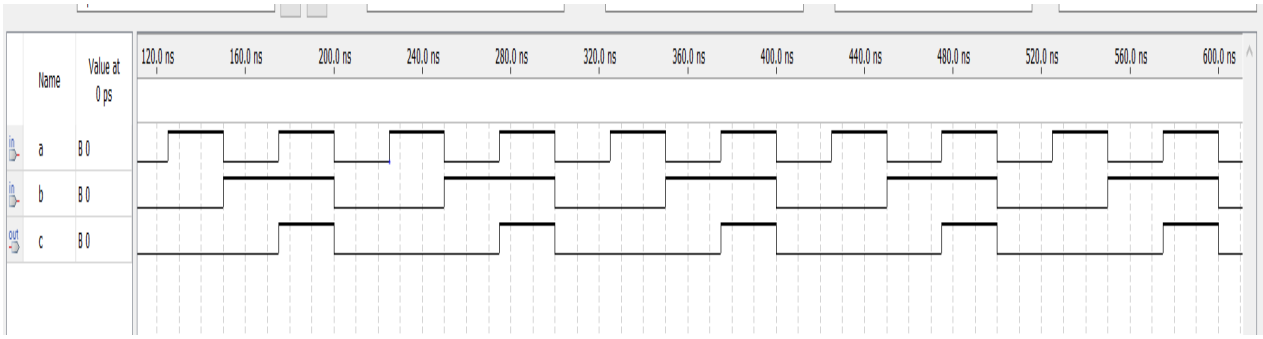
For D = 1, E = 1



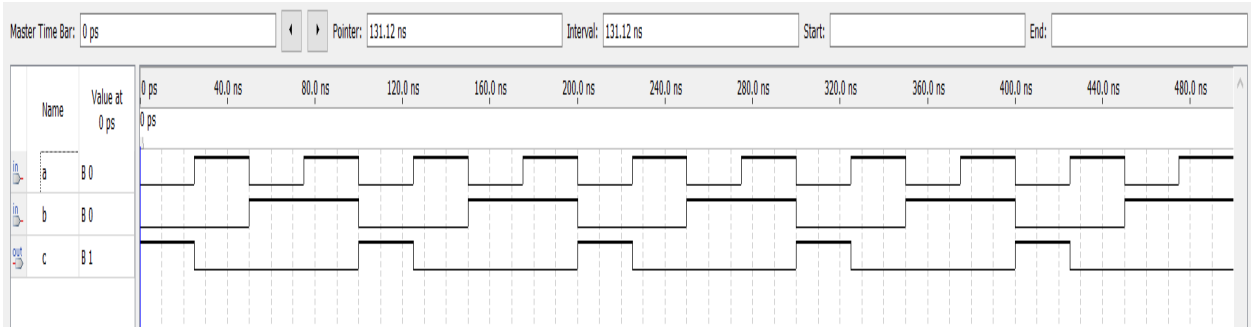
We also simulated this code via. D = 50% duty cycle 10 ns Period, E = 25% duty cycle 10 ns period signals.



Simulation (And Gate):



Simulation (Nor Gate):



Simulation (Not Gate):

